

Out-of-Sample Evaluation of Shock-Augmented Bayesian Vector Autoregressions: A G7 Macroeconomic Study

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Abstract—Bayesian vector autoregressions (BVARs) augmented with constructed exogenous shocks are widely used for macroeconomic forecasting, yet empirical practice frequently omits the safeguards needed to interpret their results. Using five quarterly U.S. series over 2000–2025, we assemble a reproducible pipeline that exposes and corrects several common failures. We show that a naïvely coded Minnesota prior can leave the posterior effectively unshrunk, yielding an explosive companion matrix (largest eigenvalue 1.068); a natural-conjugate Minnesota prior implemented through dummy observations restores stability (0.96) and bounded forecasts. We document that stacking all available shocks inflates out-of-sample error by an order of magnitude relative to a no-shock baseline, that look-ahead leakage in standardized shock construction flatters apparent skill, and that a static 13-quarter evaluation window has no power to discriminate models. Replacing it with an expanding-window evaluation (48 origins) scored by proper density rules (continuous ranked probability score, log score) and Diebold–Mariano tests shows that a stochastic-volatility BVAR produces sharper, better-calibrated predictive densities than random-walk and AR(1) benchmarks for output growth and inflation, with the long-rate block remaining hard to beat. Johansen testing returns full cointegration rank, formally justifying estimation in levels, and a diagnostic shows the COVID-19 variance spike is concentrated in output, motivating a variable-specific stochastic-volatility treatment that recovers both the Global Financial Crisis and pandemic episodes. Extending the evaluation to a five-economy G7 panel with a common global oil shock, multiple horizons, and an expanded benchmark set sharpens these conclusions: Bayesian shrinkage dominates an unrestricted VAR in every country and is the only feature a Model Confidence Set retains for most variables; the BVAR significantly outperforms the random walk for output growth and the policy rate once loss differentials are pooled across countries; and the constructed shocks improve in-sample fit but not multi-step forecasts. The study offers a transparent, reproducible evaluation protocol and a cautionary, cross-country account of where shock-augmented BVARs do and do not add value.

Index Terms—Bayesian vector autoregression, macroeconomic forecasting, Minnesota prior, stochastic volatility, density forecast evaluation, local projections, cointegration.

I. INTRODUCTION

Vector autoregressions (VARs) have been a workhorse of empirical macroeconomics since [1]. Because unrestricted VARs are heavily over-parameterized, Bayesian shrinkage through the Minnesota prior [2], [3] has become standard, and large BVARs are now routinely estimated for forecasting and

structural analysis [4], [7]. A common extension augments the endogenous block with *exogenous shocks*—series constructed outside the model (oil-price surprises, credit or fiscal impulses, monetary surprises) and entered as additional regressors or used to identify structural responses.

While the modeling toolkit is mature, applied work frequently couples it with an evaluation that cannot support the conclusions drawn. Three failures recur. First, the Minnesota prior is parameterized through precision/variance conventions that are easy to invert, so that the intended shrinkage is not actually imposed. Second, models are compared only against one another, without the naïve benchmarks that define a forecasting “horse race,” and without tests of whether accuracy differences are statistically meaningful—an acute problem given the short post-2020 evaluation windows now common. Third, shocks are standardized using full-sample moments, leaking future information into the out-of-sample period.

This paper assembles a transparent, reproducible pipeline that exposes and corrects these failures on a concrete dataset—five quarterly U.S. macroeconomic series, 2000–2025, with six constructed shocks—and reports what survives rigorous evaluation. Our contributions are:

- We give a drop-in natural-conjugate Minnesota BVAR via dummy observations [4], [5] that, on real data, converts an explosive ordinary-least-squares (OLS) VAR (largest companion eigenvalue 1.068) into a stable posterior (0.96) with bounded forecasts.
- We quantify *shock-combination overfitting*: stacking all six shocks raises out-of-sample root-mean-squared error (RMSE) by up to $8\times$ relative to a no-shock baseline, whereas the single most useful shock (oil) yields only modest gains.
- We implement an evaluation protocol with random-walk and AR(1) benchmarks, Diebold–Mariano (DM) tests with the Harvey–Leybourne–Newbold (HLN) small-sample correction [9], [10], and proper density scores [11], and show that an expanding-window design recovers the statistical power that a 13-quarter window lacks.
- We measure the effect of look-ahead leakage in shock construction, showing that the only statistically significant benchmark improvement disappears once shocks are built in real time.

TABLE I
VARIABLES.

Block	Variable	Transformation
Endogenous	GDP growth	YoY %
	Inflation (CPI)	YoY %
	REER	YoY %
	10y Treasury yield	level
Shocks	Policy rate	level
	Oil (WTI)	AR(1) surprise, z -scored
	Equity wealth (S&P 100)	AR(1) surprise, z -scored
	Consumption (PCE)	AR(1) surprise, z -scored
	Investment (PFI)	AR(1) surprise, z -scored
	Credit impulse	Δ , z -scored
	Fiscal impulse (CAPB)	Δ , z -scored

- We use Johansen cointegration testing to justify levels estimation, and a volatility diagnostic to show that the pandemic variance spike is output-specific, motivating a variable-specific stochastic-volatility (SV) model estimated with the Kim–Shephard–Chib (KSC) mixture sampler [14], [15].
- We *generalize* the headline forecasting findings to a five-economy G7 panel (United States, United Kingdom, Canada, Germany, Japan) with a common globally identified oil shock, horizons $h \in \{1, 2, 4, 8\}$, and an expanded benchmark set; pooling across countries and applying a Hansen Model Confidence Set [25] establishes which conclusions are systematic rather than U.S.-specific.

The empirical bottom line is deliberately measured: once correctly specified and honestly evaluated, the BVAR’s density forecasts beat naïve benchmarks for output growth and inflation, but the gains are concentrated, modest, and—in a short window—rarely significant; interest-rate variables are barely distinguishable from a no-change forecast.

II. DATA

We use quarterly U.S. data from 2000Q1 to 2025Q1 ($T = 101$). The endogenous block $y_t \in \mathbb{R}^5$ comprises real GDP growth (year-over-year), CPI inflation (year-over-year), the real effective exchange rate (REER, year-over-year), the 10-year Treasury yield, and a short-term policy rate. Six exogenous shocks are constructed from auxiliary series: West Texas Intermediate crude oil, an S&P 100 equity-wealth index, personal consumption expenditure, private fixed investment, nonfinancial-sector debt (a credit impulse), and the cyclically adjusted primary balance (a fiscal impulse). Table I summarizes the variables.

III. METHODOLOGY

A. BVAR with a natural-conjugate Minnesota prior

The reduced-form model with p lags and exogenous regressors x_t (a constant and the shocks) is

$$y_t = c + \sum_{j=1}^p B_j y_{t-j} + \Gamma x_t + u_t, \quad u_t \sim \mathcal{N}(0, \Sigma). \quad (1)$$

Collecting coefficients into B ($k \times n$, $k = np + m$) and stacking T observations as $Y = XB + U$, the natural-conjugate prior places $\text{vec}(B) \mid \Sigma \sim \mathcal{N}(\text{vec}(B_0), \Sigma \otimes \Omega_0)$ and $\Sigma \sim \mathcal{IW}(S_0, \nu_0)$. The Minnesota specification [2] centers each variable’s own first lag at δ_i (unity for persistent levels, below unity for mean-reverting growth rates) and all other coefficients at zero, with prior variance for the coefficient on lag ℓ of variable j in equation i proportional to $(\lambda/\ell)^2(\sigma_i/\sigma_j)^2$, where σ_i is the residual standard deviation from a univariate AR(p) and λ controls overall tightness.

We impose this prior through *dummy observations* [4], [5]. Artificial rows (Y_d, X_d) are appended so that an OLS regression on the stacked system $([Y; Y_d], [X; X_d])$ reproduces the conjugate posterior. The blocks encode (i) the own- and cross-lag prior with harmonic lag decay scaled by λ and σ_j , (ii) the inverse-Wishart scale, (iii) a finite, scale-tied prior on the exogenous coefficients (a near-flat exogenous prior allows posterior draws of shock coefficients to diverge in short windows), and (iv) a sum-of-coefficients prior, with tightness μ , that damps near-unit-root behavior. The posterior is $\Sigma \sim \mathcal{IW}(\tilde{S}, \tilde{\nu})$ and $\text{vec}(B) \mid \Sigma \sim \mathcal{N}(\text{vec}(\tilde{B}), \Sigma \otimes \tilde{\Omega})$ with $\tilde{B} = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'\tilde{Y}$ on the stacked data. Forecast stability is monitored through the largest eigenvalue of the companion matrix.

B. Shock construction and look-ahead leakage

For the four price-type series the shock is the z -scored residual of a univariate AR(1) on quarterly log growth; for credit and fiscal it is the z -scored first difference. Standardizing with full-sample mean and variance, and fitting the AR(1) on the full sample, leaks post-sample information into the shocks used during the evaluation period. We therefore construct two leak-free variants: a *train-frozen* version, in which AR and standardization parameters are estimated only through 2021Q4 and then applied out of sample, and a real-time *rolling* version using a trailing five-year window.

C. Identification: recursive SVAR and local projections

We contrast two identification philosophies with the exogenous-regressor approach. A recursive (Cholesky) structural VAR orthogonalizes the reduced-form residuals under an ordering (output, prices, exchange rate, long rate, policy rate last), yielding structural impulse responses that depend on that ordering. Local projections [12] estimate the horizon- h response by regressing y_{t+h} on the shock and controls; they are robust to VAR misspecification but require an identified shock, becoming LP-IV when the shock is used as an instrument [21], [22].

D. Forecast evaluation

We evaluate one-step-ahead forecasts in an expanding window: at each origin the model and benchmarks are re-estimated and a predictive distribution for the next quarter is formed. Benchmarks are a random walk (no-change) and an AR(1). Point accuracy is summarized by RMSE; equal predictive

TABLE II
INTEGRATION ORDER AND JOHANSEN COINTEGRATION.

Variable	ADF p	KPSS p	Verdict
GDP growth	0.006	0.10	$I(0)$
Inflation	0.968	0.05	$I(1)$
REER growth	0.454	0.10	ambiguous
10y yield	0.114	0.01	$I(1)$
Policy rate	0.002	0.10	$I(0)$
Johansen trace: rank $r = 5$ of 5 $\Rightarrow$ system stationary			

accuracy is tested by the DM statistic with the HLN small-sample correction [9], [10]. Density forecasts are scored by the log predictive score and the continuous ranked probability score (CRPS) [11],

$$\text{CRPS}(F, y) = \mathbb{E}_F |X - y| - \frac{1}{2} \mathbb{E}_F |X - X'|, \quad (2)$$

both reported as means over origins (lower is better), together with the empirical coverage of the 90% predictive interval. The BVAR predictive density is the posterior predictive ensemble; benchmark densities are Gaussian with variance estimated on the training window.

E. Stochastic volatility

To accommodate time-varying residual variance we use a univariate SV model for the relevant equation,

$$u_t = e^{h_t/2} \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1), \quad h_t = \mu + \phi(h_{t-1} - \mu) + \eta_t, \quad (3)$$

with $\eta_t \sim \mathcal{N}(0, \sigma_\eta^2)$. Estimation uses the KSC seven-component mixture approximation to $\log \varepsilon_t^2$ together with forward-filtering backward-sampling for the volatility states [14], [15]. We also considered a single *common* volatility factor [16]; Section IV-H explains why it is the wrong specification here.

IV. EMPIRICAL RESULTS

A. Integration and cointegration

Augmented Dickey–Fuller and KPSS tests classify GDP growth and the policy rate as $I(0)$, inflation and the 10-year yield as $I(1)$, and REER growth as ambiguous (Table II). The Johansen trace test returns *full* cointegration rank ($r = 5$) for the system, and full rank again ($r = 3$) for the {10y, policy rate, inflation} subsystem. Full rank means every linear combination is stationary: the system has no common stochastic trend to error-correct, a vector error-correction model collapses to a level VAR, and the explosive OLS eigenvalue is a small-sample estimation artifact rather than a unit-root pathology. Estimation in levels with Bayesian shrinkage is therefore appropriate.

B. Specification and stability

On the real data the OLS VAR is explosive: the largest companion eigenvalue is 1.068, and an unshrunk (or mis-shrunk) posterior inherits this, with multi-step forecasts diverging. The dummy-observation Minnesota BVAR with $\lambda = 0.05$ and sum-of-coefficients tightness $\mu = 0.3$ brings the largest eigenvalue

TABLE III
SHOCK-SPECIFICATION “HORSE RACE”: STATIC ONE-STEP RMSE, 2022Q1–2025Q1.

Variable	Baseline	Best single shock	All shocks
GDP growth	0.604	0.561 (credit)	4.590
Inflation	0.276	0.261 (oil)	0.691
REER	2.821	2.526 (oil)	8.277
10y yield	0.534	0.342 (oil)	1.896
Policy rate	0.544	0.270 (inv.)	0.490

to 0.96, with 78% of posterior draws stable and bounded forecasts. Tighter priors are required than textbook defaults precisely because the data are highly persistent.

C. Shock-combination overfitting

Table III reports the static one-step RMSE over 2022Q1–2025Q1 for a no-shock baseline, the single best shock per equation, and all six shocks together. Stacking all shocks is dramatically worse than using none—GDP RMSE rises from 0.60 to 4.59 and REER from 2.82 to 8.28—a textbook manifestation of over-parameterization on ~ 83 training quarters. The best single shock helps only modestly, and oil is the most consistently useful, winning three of five equations. This pattern cautions against the common instinct to add shocks cumulatively.

D. Out-of-sample point and density evaluation

A static 13-quarter window cannot discriminate models: with leak-free shocks no DM test rejects equal accuracy at the 5% level. We therefore evaluate in an expanding window from 2013Q1 (48 origins), re-estimating every model each quarter and scoring the full predictive density. Table IV reports RMSE, CRPS, log score, and 90% coverage for the SV-BVAR and the two benchmarks; Fig. 1 visualizes the CRPS comparison.

The BVAR’s predictive densities are sharper and better calibrated than the benchmarks for GDP growth and inflation on both CRPS and log score, and tie or beat the long rate on log score; coverage lies in 0.85–0.92, close to the 0.90 target. With 48 origins the DM tests acquire power: GDP beats the random walk at $p = 0.068$ and inflation beats AR(1) at $p = 0.051$, in contrast to the complete absence of significance at $n = 13$. The interest-rate block remains hard to beat—unsurprisingly, a persistent rate is well approximated by a no-change forecast.

E. Multiple horizons and an expanded benchmark set

Table V extends the evaluation to horizons $h \in \{1, 2, 4, 8\}$ and adds two benchmarks that isolate the contribution of each model component: an unrestricted (OLS) VAR and a no-shock BVAR. Multi-step forecasts are *ex ante*, with future shocks set to zero. Two results stand out. First, Bayesian shrinkage is decisive at every horizon: the unrestricted VAR is far worse than the BVAR (for GDP, CRPS of 2.27 versus 1.42 at $h = 4$), confirming that the explosive OLS dynamics translate into poor forecasts. Second, the constructed shocks do not help forecasting: the full BVAR and the no-shock BVAR

TABLE IV
EXPANDING-WINDOW ONE-STEP EVALUATION, 48 ORIGINS
(2013Q1–2025Q1). RMSE AND CRPS: LOWER IS BETTER; LOG SCORE
(LS): HIGHER IS BETTER; Cov: 90% INTERVAL COVERAGE.

Variable	Model	RMSE	CRPS	LS	Cov
GDP growth	SV-BVAR	1.785	0.685	−2.78	0.92
	RW	2.532	1.068	−3.33	0.88
	AR(1)	2.311	0.948	−3.30	0.90
Inflation	SV-BVAR	0.379	0.183	−0.49	0.90
	RW	0.407	0.201	−0.61	0.83
	AR(1)	0.439	0.211	−0.78	0.85
REER	SV-BVAR	4.321	2.117	−2.69	0.85
	RW	3.465	1.996	−2.68	0.90
	AR(1)	3.349	1.932	−2.64	0.94
10y yield	SV-BVAR	0.448	0.215	−0.44	0.88
	RW	0.383	0.216	−0.50	0.94
	AR(1)	0.386	0.217	−0.50	0.92
Policy rate	SV-BVAR	0.535	0.258	−0.57	0.88
	RW	0.443	0.229	−0.65	0.92
	AR(1)	0.460	0.236	−0.68	0.92

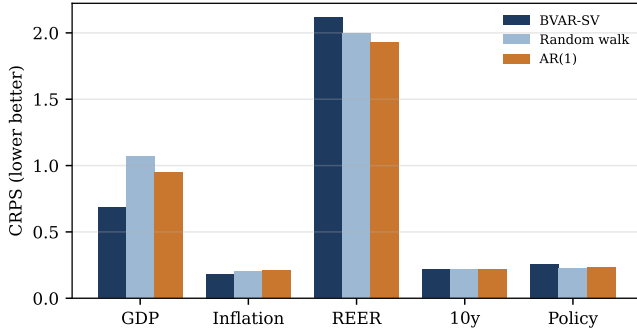


Fig. 1. Mean CRPS by variable over 48 expanding-window origins (lower is better). The SV-BVAR improves on both naïve benchmarks for GDP growth and inflation; the rate variables are close to a no-change forecast.

are essentially indistinguishable across variables and horizons. Because the shocks are observed only contemporaneously, they aid in-sample fit and identification but cannot improve genuine multi-step prediction unless the shocks themselves are forecast. A well-specified AR(1) remains a strong univariate competitor for output.

F. The cost of look-ahead leakage

Table VI compares the static one-step RMSE under the (leaky) full-sample shocks, the train-frozen shocks, and the rolling real-time shocks. The leakage is modest but systematically flattering: it lowers GDP RMSE appreciably, and—decisively—the only statistically significant benchmark improvement under leaky shocks, REER beating the random walk ($p = 0.026$), *vanishes* once shocks are built in real time ($p = 0.14$ frozen, $p = 0.18$ rolling). Apparent skill that does not survive leak-free construction should not be reported as genuine.

TABLE V
U.S. MULTI-HORIZON CRPS BY MODEL (LOWER IS BETTER). FUTURE SHOCKS SET TO ZERO (EX ANTE).

Variable	Model	$h=1$	$h=2$	$h=4$	$h=8$
GDP growth	BVAR	1.048	1.158	1.418	1.227
	BVAR no-shock	1.045	1.183	1.428	1.277
	VAR (OLS)	1.493	1.863	2.266	2.104
	RW	1.170	1.463	1.929	1.819
	AR(1)	1.036	1.173	1.418	1.269
Inflation	BVAR	0.210	0.346	0.589	0.830
	BVAR no-shock	0.207	0.341	0.594	0.872
	VAR (OLS)	0.302	0.509	1.013	1.093
	RW	0.197	0.331	0.625	1.020
	AR(1)	0.212	0.361	0.687	1.122

TABLE VI
LOOK-AHEAD LEAKAGE: STATIC ONE-STEP RMSE BY SHOCK CONSTRUCTION.

Variable	Leaky (full-sample)	Train-frozen	Rolling
GDP growth	0.478	0.526	0.618
Inflation	0.351	0.347	0.400
REER	3.136	3.172	3.183
10y yield	0.467	0.475	0.481
Policy rate	0.725	0.696	0.682

G. Structural responses

For completeness we report local-projection responses to a one-standard-deviation oil surprise (Fig. 2). Inflation rises on impact and peaks near +0.20 percentage points after two quarters, and the policy rate climbs steadily to roughly +0.37 after two years—the systematic tightening response to oil-driven inflation. The early positive output response illustrates that an unconditional oil-price surprise conflates supply and demand disturbances [18], a caution against treating it as a clean supply shock. A feasibility check found that, used as instruments in an LP-IV/proxy-SVAR, the constructed shocks are mostly weak (first-stage $F < 10$), with the strongest apparent instrument (consumption for output) invalid by construction; only oil provides a defensible, if moderate, instrument.

H. Variable-specific volatility

The constant-variance assumption fails sharply around 2020: residuals are $3.9\times$ their normal magnitude for GDP growth during 2020Q2–2021Q2, but only $1.6\times$ for inflation and essentially unchanged for the exchange rate and rates. The volatility shift is therefore *output-specific*, not common—and a single common-volatility factor, which we implemented, dilutes it and proves unidentifiable on a 95-quarter sample. A univariate SV model on the GDP residuals (Fig. 3) recovers elevated volatility at both the Global Financial Crisis ($\approx 1.6\times$ the calm level) and the pandemic ($\approx 2.7\times$), with mean-reverting persistence ($\phi \approx 0.85$). For full-system forecasting, indicator variables for the pandemic quarters [23] are a pragmatic equivalent that prevents the predictive variance from exploding; a full per-equation SV-VAR [14], [17] is the complete treatment but is marginal at quarterly frequency.

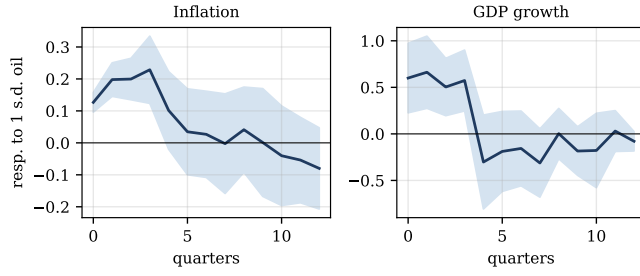


Fig. 2. Local-projection responses to a one-standard-deviation oil surprise, with 90% Newey–West bands.

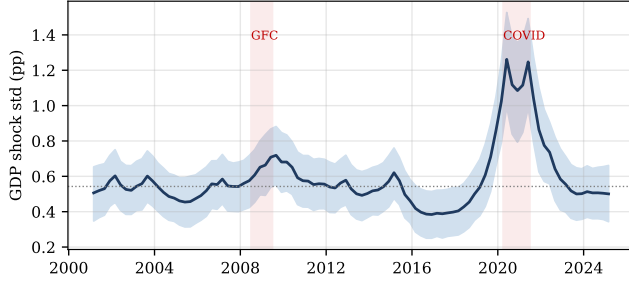


Fig. 3. Time-varying volatility of GDP-growth shocks from the univariate SV model (posterior mean and 68% band). Shaded regions mark the Global Financial Crisis and COVID-19.

V. GENERALIZATION: A G7 PANEL

The preceding analysis is a single-country case study. To establish which findings are systematic, we replicate the evaluation across five G7 economies—the United States, United Kingdom, Canada, Germany, and Japan—using comparable quarterly series from the OECD and Bank for International Settlements (real GDP growth, CPI inflation, the real effective exchange rate, the 10-year government yield, and a short/policy rate). Two design choices keep the panel clean. First, the exogenous block is a *single globally identified shock*—the oil-price surprise—which is genuinely exogenous to each economy and was the most useful shock in the U.S. analysis; this avoids reconstructing nationally specific surrogate shocks of questionable exogeneity. Second, each economy is run through the *identical* pipeline, and results are aggregated by pooling one-step loss differentials across countries for panel Diebold–Mariano tests and by computing a Hansen Model Confidence Set (MCS) [25] per variable via a stationary bootstrap. Samples are 2000–2025 for all economies except Japan (2002–2021, owing to source data limits), and heterogeneity across countries is treated as a feature.

Table VII reports, for each variable, the number of economies in which the BVAR attains lower one-step CRPS than each benchmark. The pattern is consistent with the U.S. results and sharper. Shrinkage dominates universally: the BVAR beats the unrestricted VAR in all five economies for every variable. The BVAR beats the random walk for output growth everywhere and the no-change forecast is hardest

TABLE VII

PANEL WIN COUNTS: NUMBER OF THE FIVE ECONOMIES IN WHICH THE BVAR ACHIEVES LOWER ONE-STEP CRPS THAN EACH BENCHMARK.

Variable	vs. RW	vs. AR(1)	vs. VAR(OLS)	vs. no-shock
GDP growth	5/5	3/5	5/5	1/5
Inflation	3/5	5/5	5/5	1/5
REER	4/5	3/5	5/5	4/5
10y yield	1/5	4/5	5/5	2/5
Policy rate	4/5	5/5	5/5	3/5

TABLE VIII

PANEL INFERENCE: POOLED (CROSS-COUNTRY) ONE-STEP DIEBOLD–MARIANO p -VALUE FOR BVAR VS. RW, AND THE 90% MODEL CONFIDENCE SET AMONG {BVAR, NO-SHOCK BVAR (nS), VAR(OLS), RW, AR(1)}.

Variable	Panel DM p (vs. RW)	90% MCS
GDP growth	0.016	all five
Inflation	0.922	BVAR, nS, RW
REER	0.292	BVAR, nS, RW, AR(1)
10y yield	0.523	BVAR, nS, RW, AR(1)
Policy rate	0.000	BVAR, nS

to beat for the long rate. Against AR(1) the BVAR wins systematically for inflation and the policy rate. Crucially, the full BVAR rarely beats the no-shock BVAR (e.g., 1/5 for both GDP and inflation), confirming across countries that the oil shock does not improve forecasts; the exception is the exchange rate (4/5), the variable most directly tied to oil.

Pooling loss differentials across countries restores the statistical power that no single short sample possesses (Table VIII). At the panel level the BVAR significantly outperforms the random walk for output growth ($p = 0.016$) and the policy rate ($p < 0.001$); inflation, the exchange rate, and the long yield are not separable from a no-change forecast. The Model Confidence Set tells a complementary story: the unrestricted VAR is eliminated for every variable except GDP, and for the policy rate only the two BVAR variants survive. Fig. 4 displays the underlying heterogeneity—the ratio of BVAR to random-walk CRPS in each economy—showing that the output-growth advantage holds in all five while the long-rate disadvantage is similarly broad.

VI. DISCUSSION

Three lessons generalize beyond this dataset. First, prior *implementation* matters as much as prior *choice*: a Minnesota prior coded with the wrong precision/variance convention silently fails to shrink, and dummy observations are a robust way to avoid this. Second, evaluation design dominates point estimates. A short window and model-versus-model comparison can manufacture apparent skill; naïve benchmarks, density scoring, and an expanding window are necessary to separate signal from noise, and on macro data the honest verdict is often “competitive but rarely significant.” Third, diagnostics should precede machinery: Johansen testing showed that a vector error-correction model was unnecessary, and a variance

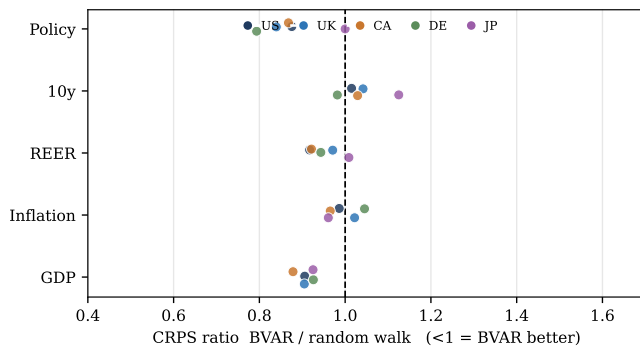


Fig. 4. One-step CRPS of the BVAR relative to the random walk, by economy and variable (values below the dashed line indicate the BVAR is better). The output-growth advantage is systematic across the G7; the long rate is uniformly hard to beat.

diagnostic showed that common stochastic volatility was the wrong tool, redirecting effort to a variable-specific treatment.

Limitations remain. Although the panel addresses the single-country concern, each economy still offers only a short quarterly sample (~ 100 observations or fewer), which limits the power of forecast comparisons and richly parameterized volatility models even after cross-country pooling; we use final-vintage rather than real-time data; the panel relies on a single exogenous shock; and the country-specific demand-side surrogates in the U.S. analysis have questionable exogeneity. Monthly data and externally identified shocks (high-frequency monetary surprises, the excess bond premium [19], [20]) are natural extensions that would make per-equation SV and proxy-SVAR identification more credible across the panel.

VII. CONCLUSION

We have presented a transparent, reproducible pipeline for shock-augmented BVAR forecasting and used it to expose and correct several common failures: mis-specified Minnesota shrinkage, shock-combination overfitting, the absence of naïve benchmarks and significance testing, look-ahead leakage in shock construction, and an inappropriate common-volatility assumption. Once these are addressed, a correctly specified stochastic-volatility BVAR delivers sharper, better-calibrated density forecasts than random-walk and AR(1) benchmarks for output growth and inflation, while interest rates remain difficult to beat. Extending the study to a five-economy G7 panel shows these conclusions are systematic rather than U.S.-specific: Bayesian shrinkage dominates an unrestricted VAR in every country, the BVAR significantly beats the random walk for output growth and the policy rate when evidence is pooled across countries, and the constructed shocks improve fit but not multi-step forecasts everywhere. The contribution is less a new model than a disciplined, cross-country protocol for deciding when a shock-augmented BVAR genuinely adds forecasting value.

REPRODUCIBILITY

All estimation, evaluation, and figure code, together with the corrected BVAR, leak-free shock construction, the cointegration and identification diagnostics, the expanding-window density evaluation, and the SV sampler, are provided as self-contained Python modules.

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